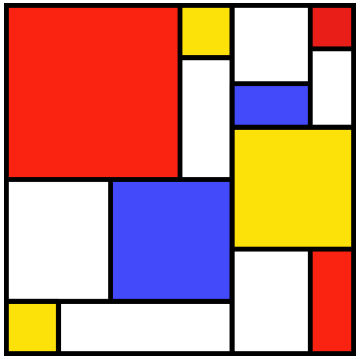
SLiM

Workshop Series



Tree-sequence Recording

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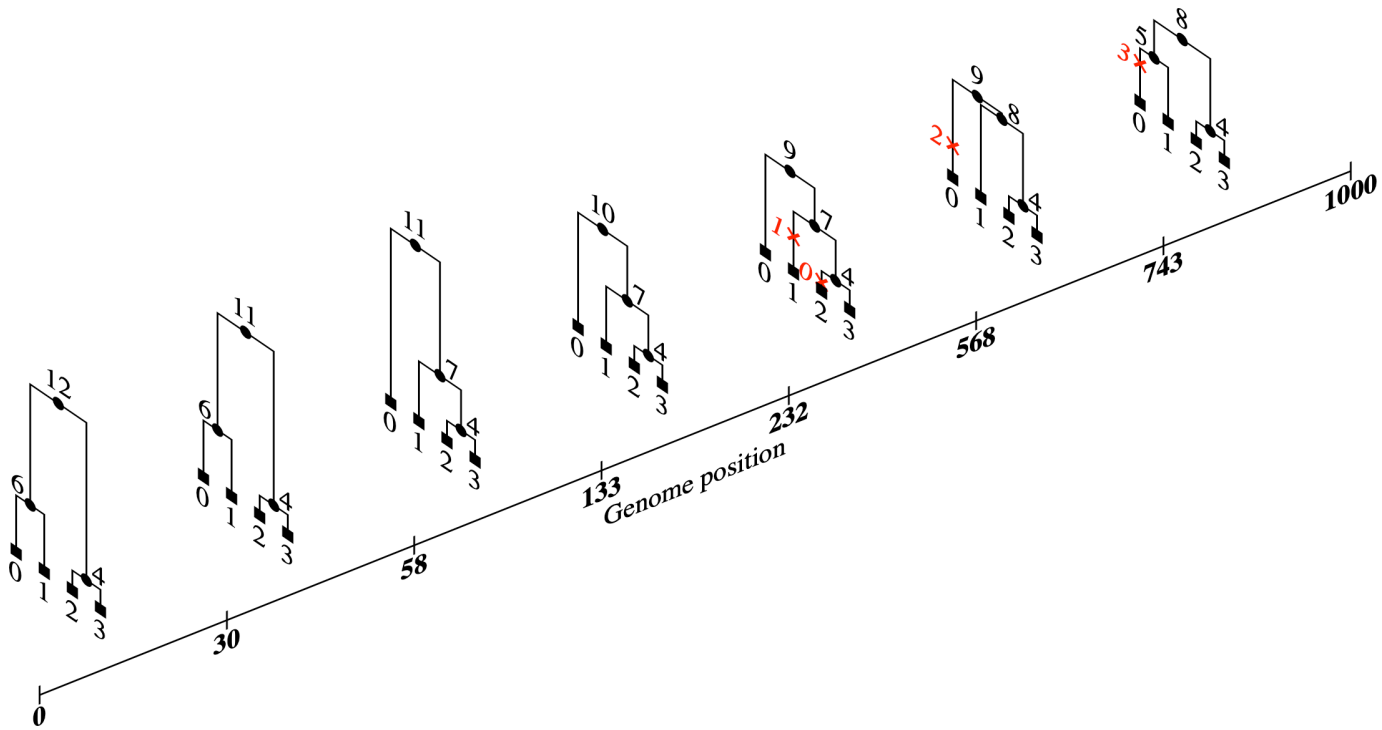
Messer Lab, Cornell University

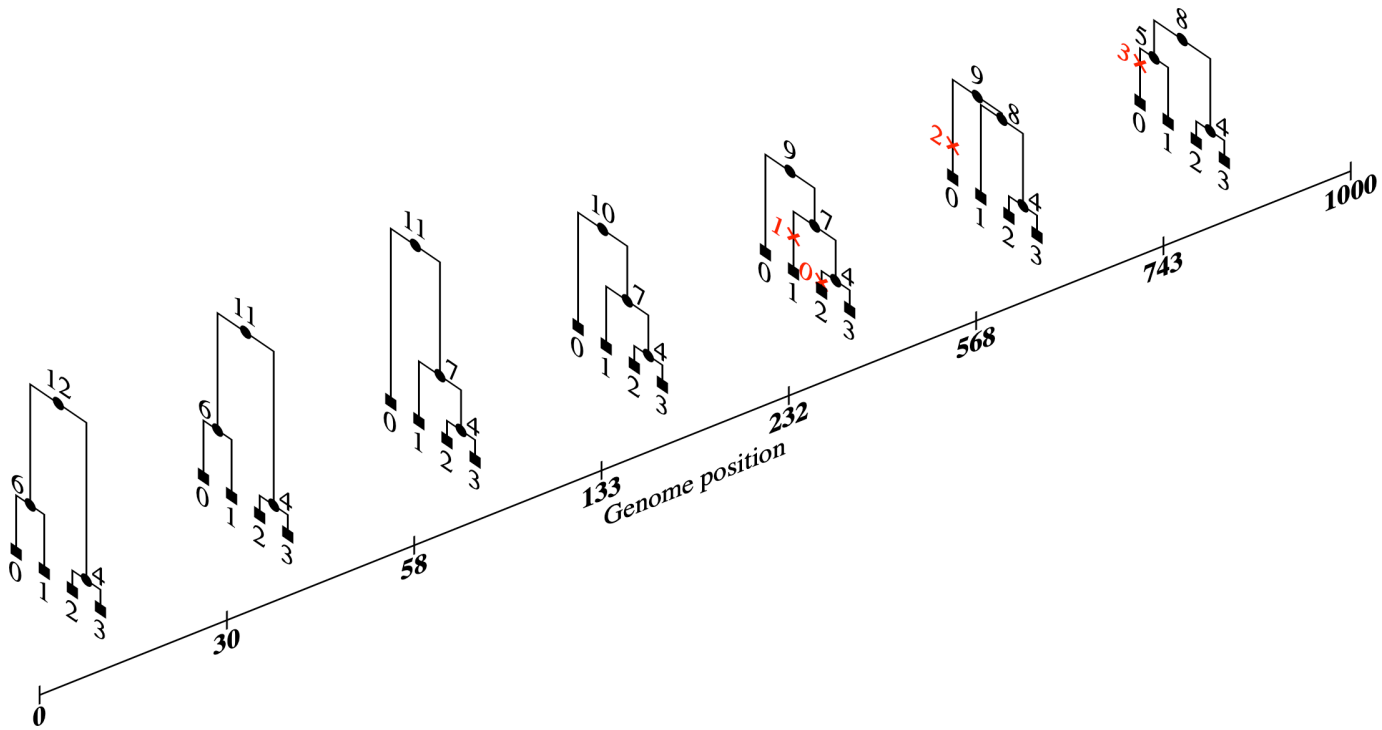
Tree sequences

- A record of the ancestry at every position
  - **Leaves** are often extant individuals
  - **Internal nodes** are ancestors
  - In SLiM, **roots** are the first generation



Tree sequences

- A record of the ancestry at every position
  - From coalescent modeling (`msprime`)
  - **Extremely compact** due to correlations
  - **Very fast** to traverse and calculate statistics



Tree-sequence recording

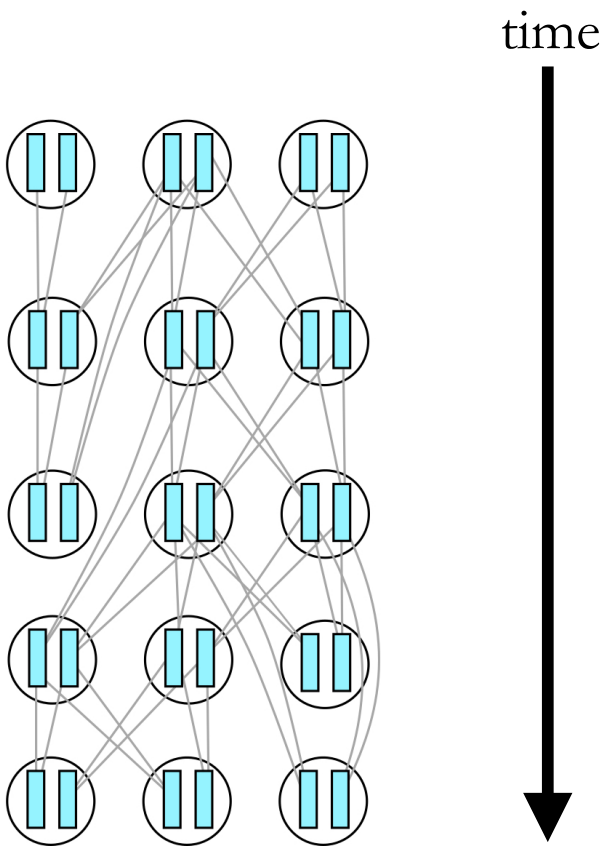
- What is the benefit of this in SLiM?
  - **Complete ancestry information**
  - **Speed** from neutral mutation overlay
  - **Speed** from recapitation for burn-in

A complete tree-seq model

```
initialize() {
    initializeTreeSeq();
    initializeMutationRate(0);
    initializeMutationType("m1", 0.5, "f", 0.0);
    initializeGenomicElementType("g1", m1, 1.0);
    initializeGenomicElement(g1, 0, 1e8-1);
    initializeRecombinationRate(1e-8);
}
1 early() {
    sim.addSubpop("p1", 500);
}
5000 late() {
    sim.treeSeqOutput("final.trees");
}
```

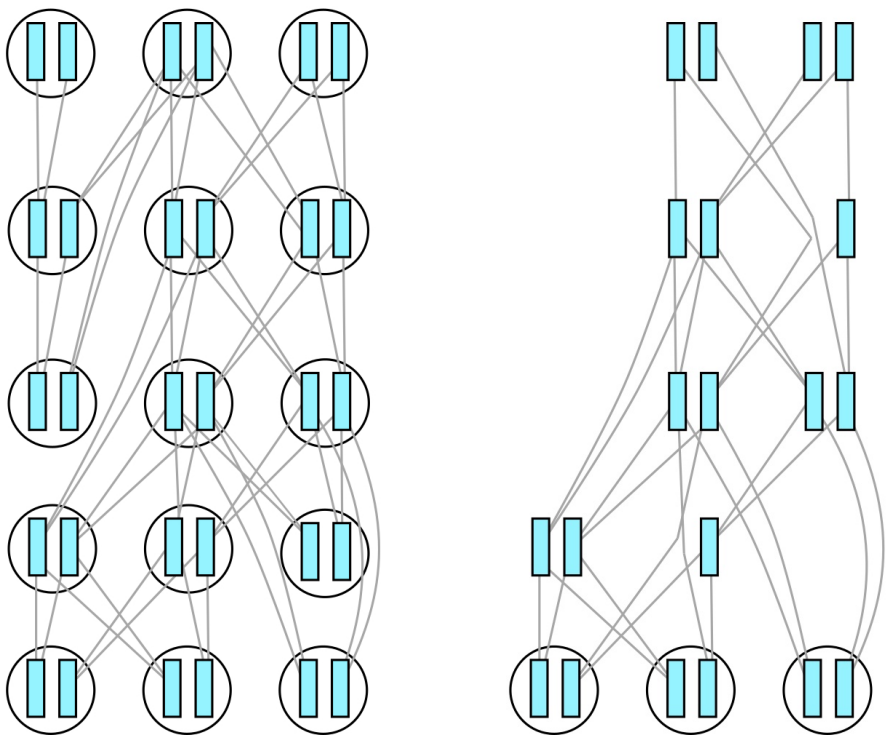
- Calls `initializeTreeSeq()` and `treeSeqOutput()`
- Uses a (neutral) mutation rate of zero

Tree-sequence recording



SLiM simulation

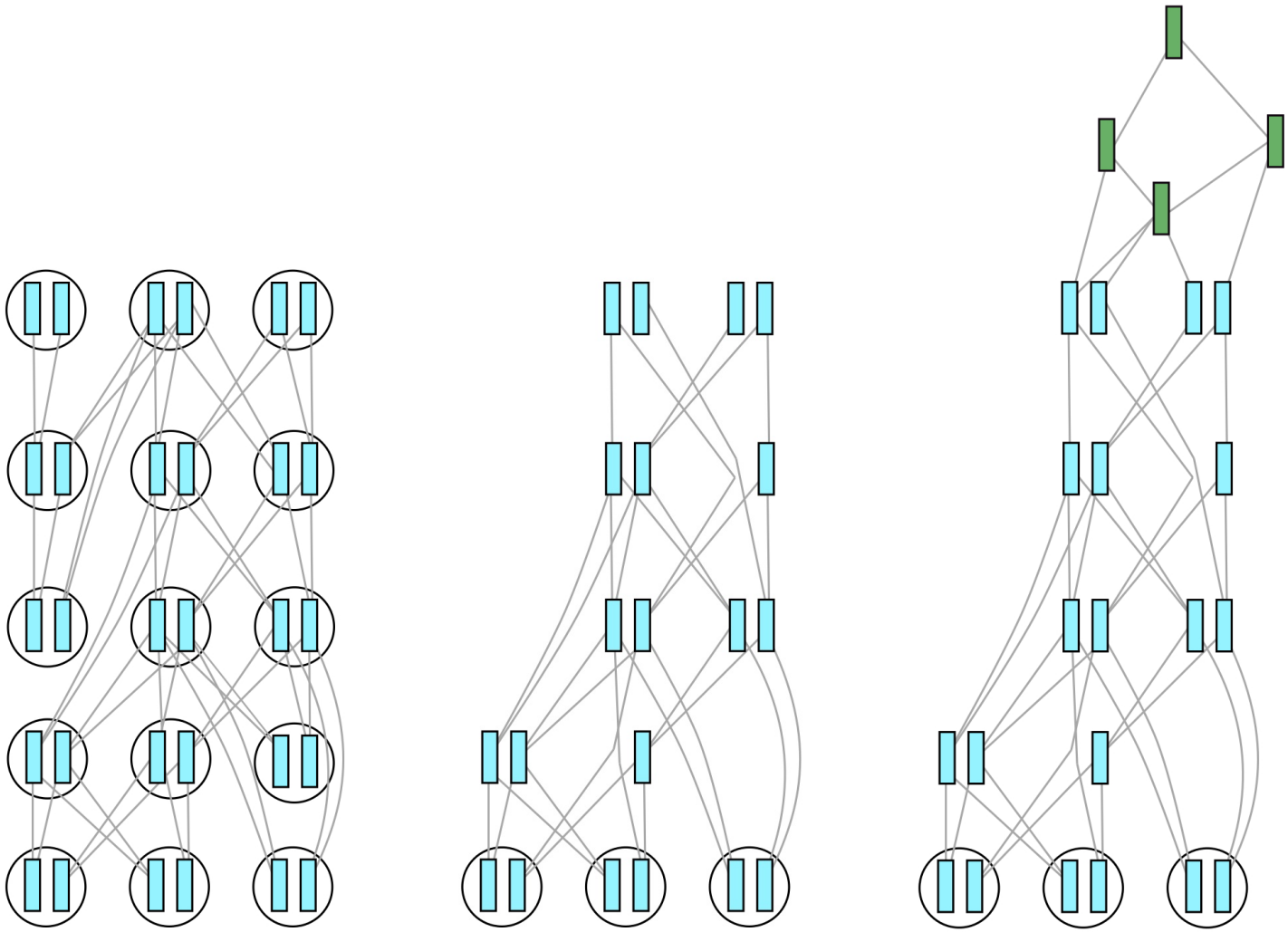
Tree-sequence recording



SLiM simulation

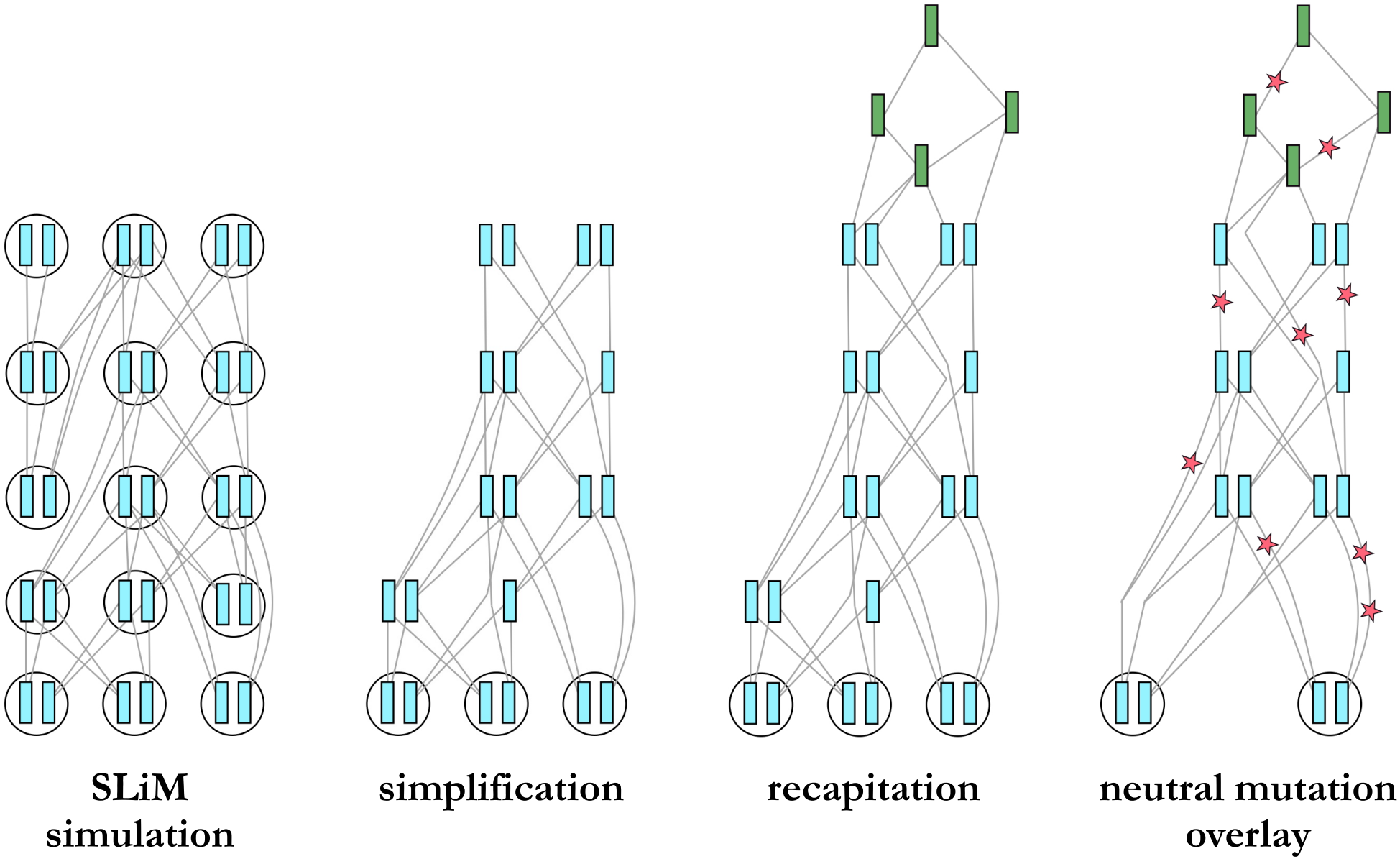
simplification

Tree-sequence recording



SLiM simulation **simplification** **recapitation**

Tree-sequence recording



Tree-sequence recording

- Enable tree-sequence recording
  - `initializeTreeSeq()`
- Control simplification if desired
  - `simplificationRatio, simplificationInterval`
  - `treeSeqSimplify()`
- Remember particular individuals if desired
  - `treeSeqRememberIndividuals()`
- Output a `.trees` file at completion
  - `treeSeqOutput()`

Tree-sequence analysis in Python

- SLiM:
  - runs forward genetic simulations
- `tskit`:
  - provides the foundation for tree sequences
- `msprime`:
  - performs mutation and coalescence
- `pyslim`:
  - knows about SLiM `.trees` files specifically

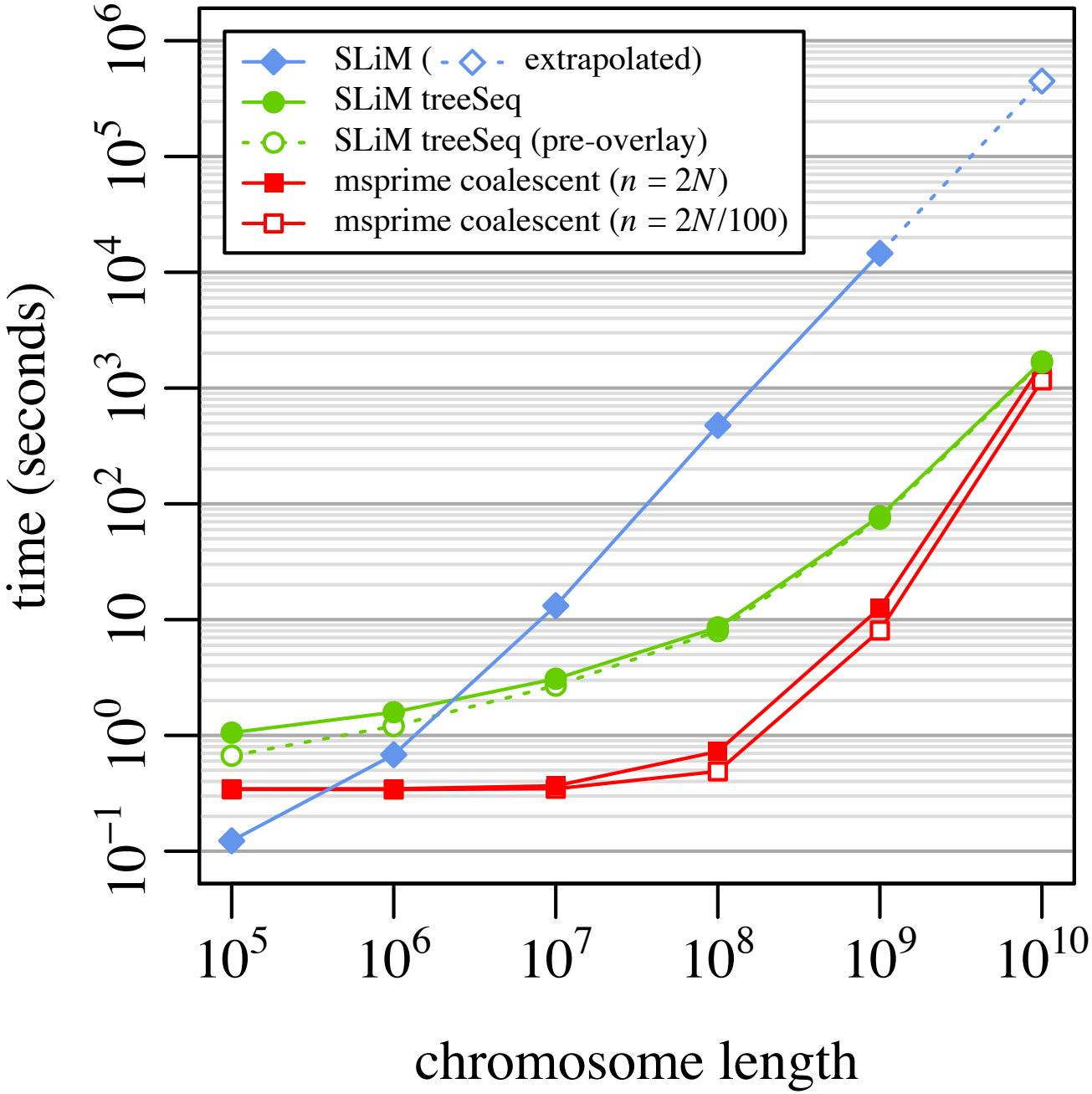
Tree-sequence analysis in Python

- Typical workflow:
  - run a simulation in SLiM and save a `.trees` file
  - read the `.trees` file from SLiM with `tskit`
  - recapitate it with `pyslim`
  - overlay mutations with `msprime`
  - perform analyses upon it in Python
  - write out a modified `.trees` file

A complete Python analysis script

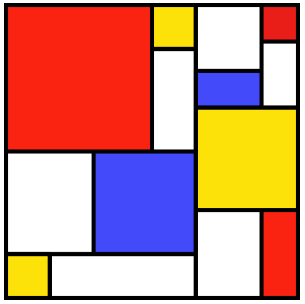
```
import msprime, tskit, pyslim

ts = tskit.load("final.trees")
recap = pyslim.recapitate(ts, ancestral_Ne=1e5,
    recombination_rate=3e-10)
mutated = msprime.sim_mutations(recap, rate=1e-7, keep=True)
mutated.dump("final_overlaid.trees")
```

- Import `msprime`, `tskit`, and `pyslim`
- Load the saved `.trees` file with `tskit`
- Use `pyslim` to recapitate it
- Use `msprime` to overlay mutations
- Write out the final tree sequence



Haller et al. (2019), Molecular Ecology Resources

Guest Lecture!



ts kit ms prime

A Practical Introduction to the Tree Sequence

Peter Ralph, University of Oregon

video: https://bit.ly/pr_tseq1

slides: https://bit.ly/pr_tseq2

Resources

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RESOURCE ARTICLE

WILEY MOLECULAR ECOLOGY RESOURCES

Tree-sequence recording in SLiM opens new horizons for forward-time simulation of whole genomes

Benjamin C. Haller[1] | Jared Galloway[2] | Jerome Kelleher[3] |
Philipp W. Messer[1,*] | Peter L. Ralph[2,*]

PLOS | COMPUTATIONAL BIOLOGY

RESEARCH ARTICLE

Efficient pedigree recording for fast population genetics simulation

Jerome Kelleher[1], Kevin R. Thornton[2], Jaime Ashander[3], Peter L. Ralph[4]*

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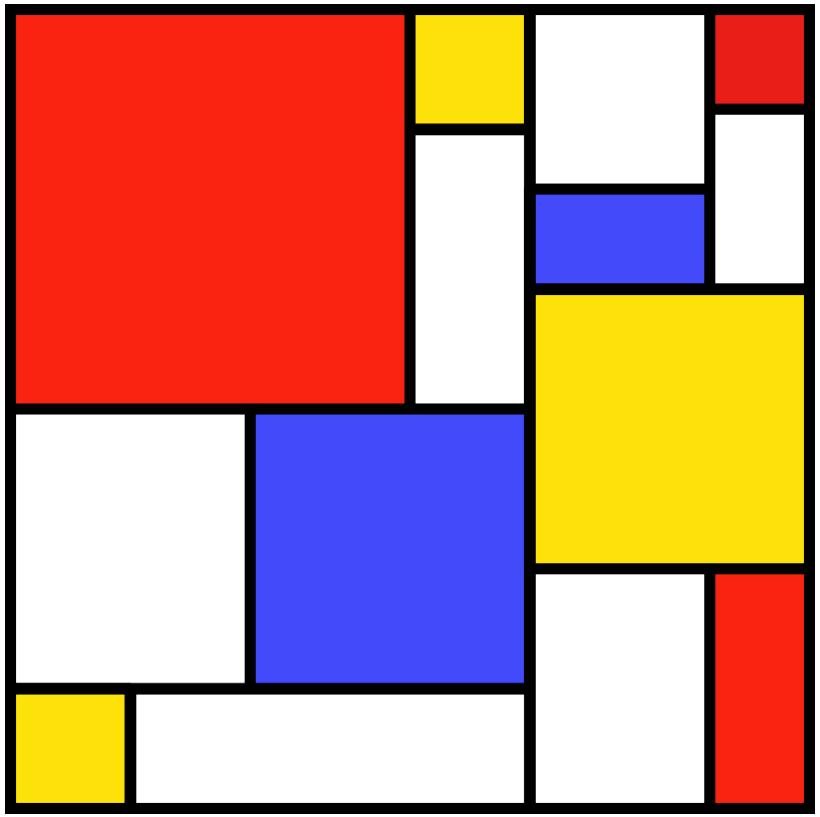

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